

Web Programming

Rodrigo Martins Pagliares

Last update: 2021/06/02

Web Programming

Rodrigo Martins Pagliares

Web programming foundations

Core web languages

The three core web languages

Lecture: The three core web languages

Module: Core web languages

Responsibilities

Structure



Design



Behavior



HTML, CSS and JavaScript

- **HTML** is the **universal document language** for the **world wide web**.
- An **HTML document** contains all the **contents** of the **page** and **references to media items** like **images, audio** and **video**.
 - ✓ It also contains **links** to **Cascading Style Sheets (CSS)** and possibly **JavaScript**.
- **CSS** contains **instructions** on **how** the **content** should be **displayed** in the **browser**.
- **JavaScript** is used for **small programs** that **run** in the **browser**.

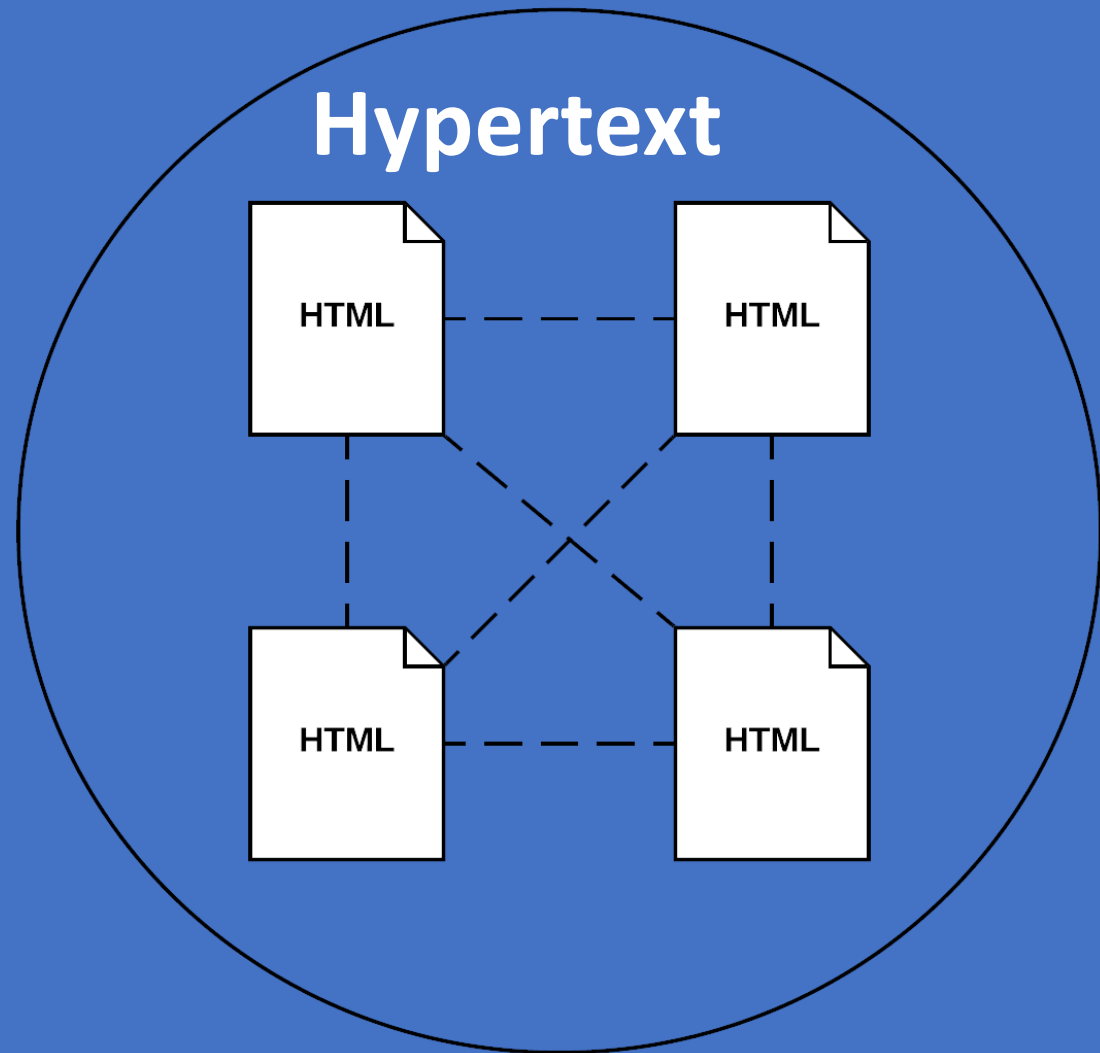
HTML

Lecture: HTML

Module: Core web languages

HTML stands for HyperText Markup Language.

Hypertext means that documents on the **World Wide Web** are **not** meant to be consumed in a linear fashion, but instead, designed to be linked to other documents.



Roles of a markup language

- A markup language defines the role each piece of content will have.
 - ✓ Headers
 - ✓ Navigation
 - ✓ Footers
 - ✓ ...
- Some **tags** describe **non-textual elements** like **images** or **video**.

HTML

- **HTML is a declarative markup language.**
 - ✓ This means that **HTML describes the contents of a document.**
- **Example**
 1. First-level heading.
 2. Paragraph.
 3. Link.
 4. Emphasized text.
 5. Blockquote.
 6. Image.
 - ✓ ...

```
<tag1>text</tag1>
```

HTML is a **markup language** consisting of a set of **elements (tags)** that are **added** to the **text** of a **webpage** to **describe** its **content** and **define** its **structure**.

```
<tag2>text</tag2>
```

Hello, cruel world with HTML!

In this course we are going to study the 3 core web languages.

HTML

CSS

JavaScript

This is an example of a basic HTML page with simple HTML elements.

Content

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world</title>
</head>
<body>
  <h1>Hello, cruel world with HTML!</h1>
  <p>In this course we are going to study the 3 core web languages.</p>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
  </ul>
  <p>This is an example of a basic HTML page with simple HTML elements.</p>
</body>
</html>
```

Markup

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world</title>
</head>
<body>
  <h1>Hello, cruel world with HTML!</h1>
  <p>In this course we are going to study the 3 core web languages.</p>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
  </ul>
  <p>This is an example of a basic HTML page with simple HTML elements.</p>
</body>
</html>
```

Human-readable Markup Content

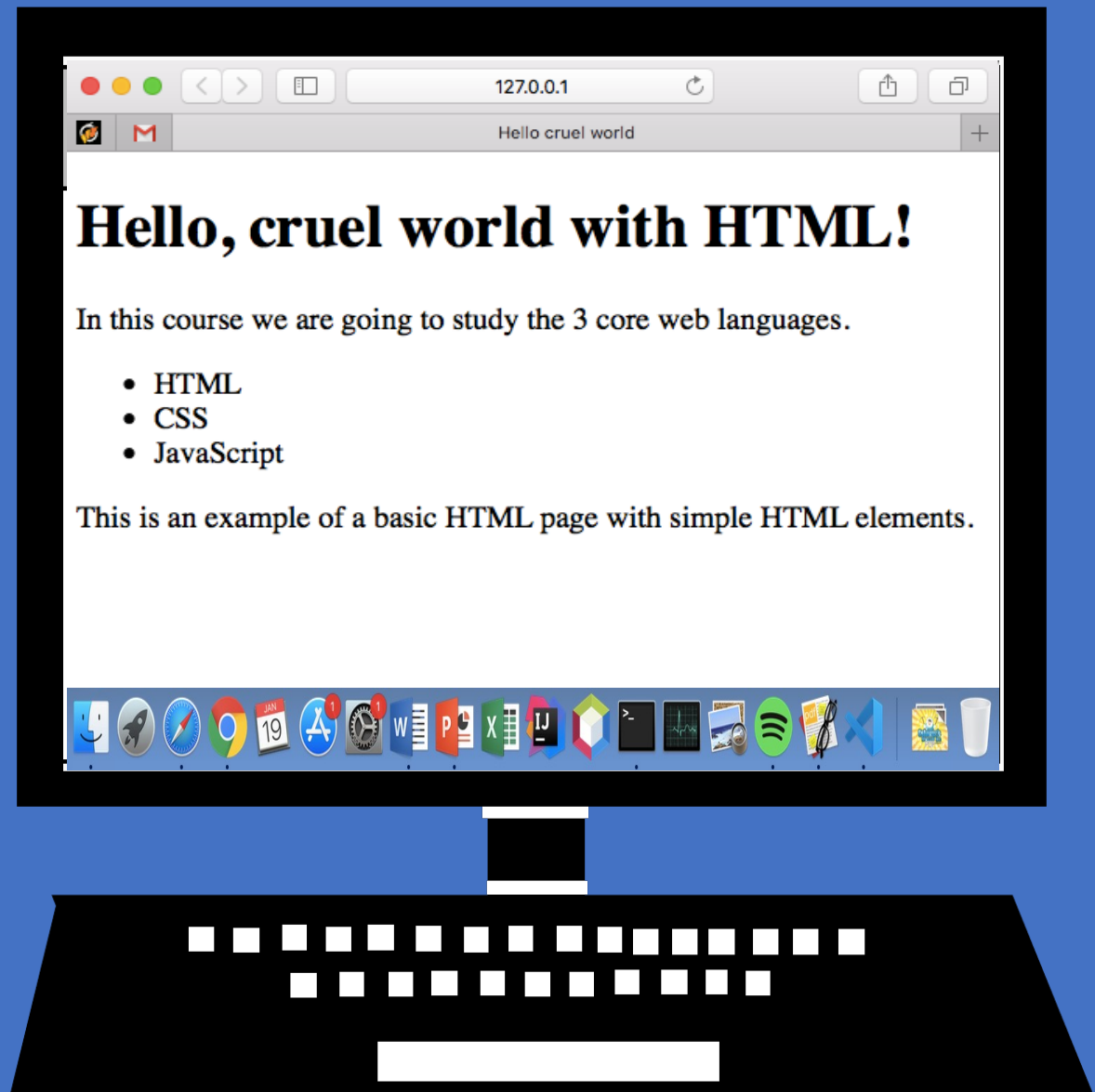

```
<!DOCTYPE html><html><head><title>Hello cruel world</title> </head><body>  
<h1>Hello, cruel world with HTML!</h1><p>In this course we are going to study  
the 3 core web languages.</p><ul><li>HTML</li><li>CSS</li><li>JavaScript</li>  
</ul><p>This is an example of a basic HTML page with simple HTML elements.</p>  
</body></html>
```

Computer-readable Markup Content

Computer-readable because a **computer** understand the **relationships** **even** if we **removed** all the **line breaks** and **spaces**, but that **would** make the **document unreadable** for **humans**.

**Content and markup provide semantic meaning
about web documents to web browsers...**

Markup
content
rendered by
a web
browser



Whether you're a **front-end, back-end,**
or **full-stack developer**, **HTML** is the language
that you **need to master**.

CSS

Lecture: CSS

Module: Core web languages

Cascading Style Sheets (CSS) is a **presentational markup language** that provides **instructions** on how **elements** in a **HTML document** should be **styled** in the **viewport**.

CSS

- **CCS** is used to **give the browser instructions about the appearance of HTML elements.**
- **CSS** creates a **clear separation** between the **content, HTML markup,** and its **presentation, CSS rules.**
- **CSS** defines a **set of rules** that **identify** the **part** of the **page** that **you want to modify,** and **then describes** the **styles** that **make the page look and behave.**

CSS history

- CSS was a **response** to an **early trend** of using **tables, frames and layers** to **force layouts** on a **vertical document** by **building presentational elements** right into the **HTML**.
 - ✓ **Mixing the content** with **things** that were **not meaningful** and **messing up** the **content** .

Web browsers have default predefined CSS rules applied to standard HTML documents.

CSS

- **When you open an unstyled HTML document in a browser:**
 - ✓ **Headings are big and bold.**
 - ✓ **Emphasized text is italicized.**
 - ✓ **Strongly emphasized text is bold.**
 - ✓ **Links are blue with an underline.**
 - ✓ **Every element has its own unique visual appearance.**

Any **CSS rule or property** we define will **override** the **default web browser rules**.

Responsive content & design

- One of the **big differences** in designing for the **web** is that **content has to be responsive**.
 - ✓ Adjust to the **proportions of different viewports**, like **desktops, laptops, and mobile devices**.



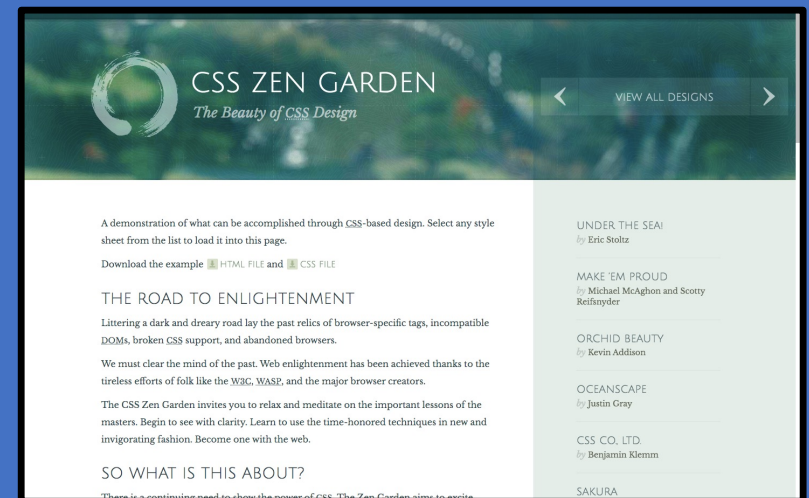
Image source: <https://bit.ly/2uEvVj1>

Design for a **website** will require **learning** how to work with the **layout language of the web**, which is called **CSS**.

CSS Zen Garden

CSS Zen Garden

- Demonstrates how the same content can be designed to look completely different.
- The site showcases different styles for the same HTML page.
 - ✓ Exactly the same HTML code, just presented in dramatically different ways.





CSS ZEN GARDEN

The Beauty of CSS Design



[VIEW ALL DESIGNS](#)



A demonstration of what can be accomplished through CSS-based design. Select any style sheet from the list to load it into this page.

Download the example  HTML FILE and  CSS FILE

THE ROAD TO ENLIGHTENMENT

Littering a dark and dreary road lay the past relics of browser-specific tags, incompatible DOMs, broken CSS support, and abandoned browsers.

We must clear the mind of the past. Web enlightenment has been achieved thanks to the tireless efforts of folk like the W3C, WASP, and the major browser creators.

The CSS Zen Garden invites you to relax and meditate on the important lessons of the masters. Begin to see with clarity. Learn to use the time-honored techniques in new and

UNDER THE SEA!

by Eric Stoltz

MAKE 'EM PROUD

by Michael McAghon and Scotty Reifsnyder

ORCHID BEAUTY

by Kevin Addison

OCEANSCAPE

by Justin Gray

CSS CO, LTD.

by Benjamin Klemm

SAKURA

<http://www.csszengarden.com>

There is a continuing need to show the power of CSS. The Zen Garden aims to excite

CSS ZEN GARDEN

THE BEAUTY OF CSS DESIGN

SELECT A DESIGN:

Mid Century Modern by

Garments by

Steel by

Apothecary by

Screen Filler by

Fountain Kiss by

A Robot Named Jimmy by

Verde Moderna by

Andrew

Dan Mall

Steffen Noel

Trent V

Elliot Jay Stoc

Jerem

meltmedia

Dave Sl

ARCHIVES:

Next Designs >

View All Designs

RESOURCES:

View This Design's
CSS

CSS Resources

FAQ

Submit a Design

Translations

Download the example
html file and css file

SO WHAT IS THIS ABOUT?

There is a continuing need to show the power of CSS. The Zen Garden aims to excite, inspire, and encourage participation. To begin, view some of the existing designs in the list. Clicking on any one will load the style sheet into this very page. The HTML remains the same, the only thing that has changed is the external CSS file. Yes, really.

5

3

CC

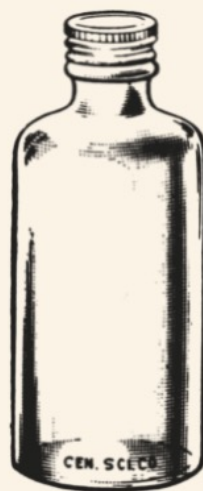
Person icon

GitHub icon

<http://www.csszengarden.com>

~ DR. SHEA'S MIRACULOUS ~

CSS Zen Garden



<http://www.csszengarden.com>

THE BIRTH OF CSS DESIGN

CSS evolution

- CSS also let you add some **motion** and **even** a level of **interactivity** to your **designs**.
- As **CSS** has **matured**, **variables**, **calculations**, and **more complex features** have made it into the **language**.
 - ✓ CSS now has some **features** that are **similar** to what's available in **programming languages**.

CSS evolution

- The **CSS language** is **always changing**.
 - ✓ There have been **three major revisions**.
- The **third revision** has been **split** into **independent modules** that will **expand** in the **future**.
 - ✓ The **language** will **always** be **evolving**.

CSS evolution

- The **switch** from **table**, **frames** and **layer** spaced **layouts** to **pure CSS layouts** meant **people** could **choose how** they wanted to **access HTML documents**.
 - ✓ Either with or without the **styles** the **designer** and **developer** has **applied**.
- A **text to speech browser**, for example, will largely **ignore CSS**, since **visual presentation does not translate** into **audio**.

CSS

- **When we supply our own CSS, our CSS rules override what the browser provides.**
- If you **want all headings h1** to be **red** and all paragraphs in **HTML** document to be **black**, we can apply the following **rules**:

```
1  h1 {  
2      color: red;  
3      font-size: 5em;  
4  }  
5  
6  p {  
7      color: black;  
8  }
```

Cascade

- **CSS works on a cascade, meaning style rules and properties are applied literally from the top down.**
 - ✓ We start with **default browser styles** and then styles are applied starting at the **top** of the **document** and moving **down**.
- In other words, **style rules further down in a CSS file have more weight than the rules higher up.**

Cascade

- This is how **CSS adds a presentational layer to web documents.**
 - ✓ **Starting from the default browser styles and targeting first the most general element and then moving in to the most specific ones, we create rules to describe their appearance.**

Less is more

- The **trick** is to **figure out how to make as few rules and property declarations as possible to create the most impact.**
- In **CSS**, **smaller and simpler** is always **better**.

JavaScript

Lecture: JavaScript

Module: Core web languages

JavaScript

- **Everyone who works in the world of web development has to be comfortable with a certain amount of programming.**
- **For web development that means learning about JavaScript.**
 - ✓ The language responsible for **managing interaction** within a **browser**.
- **Stack Overflow 2019 survey:**
 - ✓ **50% of back-end developers** say they use some form of **JavaScript daily**.

Some definitions of JavaScript ...

JavaScript is an **interactive layer** that **sits on top** of
HTML and **CSS**.

Source: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>

JavaScript is a **scripting language** that **allows** you to **write** **small programs** that **run** in the **browser** and **change** the **HTML** and **CSS** of the **current document** to **change** what you **see** and what you **can do**.

Source: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>

Why do we use JavaScript?

JavaScript is used to **interact** with **HTML**, **CSS**, and **events**
within the **browser viewport**.

JavaScript

- JavaScript loads into the **browser** to add **interactive behaviors**.
- **Examples of interactivity:**
 - ✓ **Hiding-showing** behavior.
 - ✓ **Sliding** animation.
 - ✓ **Loading all new content** into the **document**.
 - ✓ **Changing** the entire **view**.

“Unplugged” JavaScript

- JavaScript interactivity happens locally on whatever device is accessing the document.
 - ✓ In many cases, JavaScript can run even if the Internet and the connection to the web server is cut.

How does JavaScript work?

When you point your browser at an address with an HTML document, the browser:

- 1. Pulls down the HTML document and indexes all its contents.**
- 2. Downloads any CSS and JavaScript documents referenced.**
- 3. Runs the JavaScript to see if any of the HTML markup has changed.**
- 4. Applies the CSS to make everything look the way it's supposed to.**
- 5. Leaves the JavaScript running in case an interaction or event triggers further scripts.**

There have been **several revisions** to the **language** with a **new version** always on the **horizon**.

JavaScript evolution

- **Traditional JavaScript:**

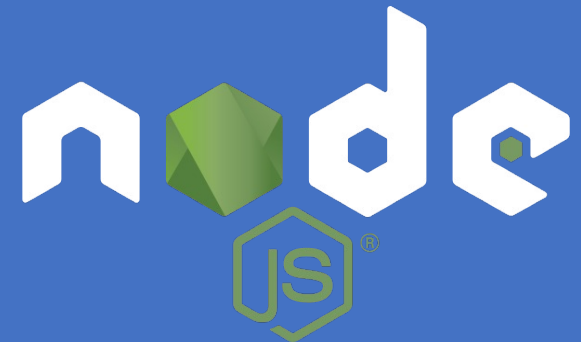
- ✓ Runs in the **web browser**, managing **HTML** and **CSS** in the **browser viewport** to **add interactivity**.

- **Modern JavaScript:**

- ✓ Runs in the **browser** (**front-end**) or **server** (**back-end**) to **generate dynamic documents** and **views**.

Back-end JavaScript

- **JavaScript** has **migrated** from the **browser** to **other environments**.
 - ✓ You can now create **server-side JavaScript applications** and **populate sites almost entirely** using **JavaScript**.





[HOME](#) | [ABOUT](#) | [DOWNLOADS](#) | [DOCS](#) | [GET INVOLVED](#) | [SECURITY](#) | [CERTIFICATION](#) | [NEWS](#)



Node.js® is a JavaScript runtime built on [Chrome's V8 JavaScript engine](#).

Join us at [OpenJS World](#), a free virtual event on June 2-3, 2021

Download for macOS (x64)

14.17.0 LTS

Recommended For Most Users

16.2.0 Current

Latest Features

[Other Downloads](#) | [Changelog](#) | [API Docs](#) [Other Downloads](#) | [Changelog](#) | [API Docs](#)

Or have a look at the [Long Term Support \(LTS\) schedule](#).



[Report Node.js issue](#) | [Report website issue](#) | [Get Help](#)

© OpenJS Foundation. All Rights Reserved. Portions of this site originally © Joyent.

Node.js is a trademark of Joyent, Inc. and is used with its permission. Please review the [Trademark List and Trademark Guidelines of the OpenJS Foundation](#).

[Node.js Project Licensing Information](#).

<http://www.nodejs.org>



Thank you dnlp for being a Node.js contributor 54 contributions

Back-end JavaScript

- Express is a minimal Node.js web application framework that provides a set of features for web and mobile applications.

Express.js

Black Lives Matter.
Support the Equal Justice Initiative.

Express

search

Home Getting started Guide API reference Advanced topics Resources

Express^{4.17.1}

Fast, unopinionated, minimalist web
framework for [Node.js](#)

```
$ npm install express --save
```

 **Express 5.0 alpha documentation is now available.**

The alpha [API documentation](#) is a work in progress. For information on what's in the release, see the Express [release history](#).

Web Applications

Express is a minimal and flexible
Node.js web application

APIs

With a myriad of HTTP utility
methods and middleware at
your disposal, Express makes it

Performance

Express provides a thin layer of
fundamental web application
features, without obscuring
Node.js features that you know
and love.

Frameworks

Many [popular frameworks](#) are
based on Express.

<http://www.expressjs.org>

Questions?